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**Batch - 15**

### **1. Algorithm**

1. Start.
2. Input a **positive integer n**.
3. If  $n \leq 0$ , display an error message and stop.
4. Display the menu:
  - 1 – Check whether all digits are unique.
  - 2 – Find the frequency of digits 0 to 9.
5. Input the user's **choice**.
6. Copy  $n$  into **temp**.
7. Use a **switch** statement.

#### **Case 1 – Check unique digits**

1. Initialize  $\text{freq}[10] = \{0\}$  and  $\text{unique} = 1$ .
2. Repeat while  $\text{temp} > 0$ :
  - Find last digit:  $\text{digit} = \text{temp} \% 10$ .
  - Increment  $\text{freq}[\text{digit}]$ .
  - If  $\text{freq}[\text{digit}] > 1$ , set  $\text{unique} = 0$  and stop checking.
  - Remove the last digit using  $\text{temp} = \text{temp} / 10$ .
3. If  $\text{unique} == 1$ , display "**Unique**".
4. Otherwise, display "**Not Unique**".

#### **Case 2 – Find frequency of each digit**

1. Initialize  $\text{freq}[10] = \{0\}$ .
2. Repeat while  $\text{temp} > 0$ :
  - Find last digit using  $\text{digit} = \text{temp} \% 10$ .
  - Increment  $\text{freq}[\text{digit}]$ .
  - Remove the last digit using  $\text{temp} = \text{temp} / 10$ .
3. Use a loop from 0 to 9 to display the frequency of every digit.

**Default:** Display "**Invalid choice**".

8. Stop.

## 2. Test Case Table

| Test Case | Input(s)                       | Expected Output(s)                                  | Actual Output(s)                                    | Result |
|-----------|--------------------------------|-----------------------------------------------------|-----------------------------------------------------|--------|
| 1         | Number = 12345,                | Unique                                              | Unique                                              | Pass   |
| 2         | Number = 12234,                | Not Unique                                          | Not Unique                                          | Pass   |
| 3         | Number = 9876543210,           | Unique                                              | Unique                                              | Pass   |
| 4         | Number = 112333,<br>Choice = 2 | freq_0=0, freq_1=2, freq_2=1,<br>freq_3=3, others=0 | freq_0=0, freq_1=2, freq_2=1,<br>freq_3=3, others=0 | Pass   |
| 5         | Number = 505050,               | freq_0=3, freq_5=3, others=0                        | freq_0=3, freq_5=3, others=0                        | Pass   |

## 3. C Program

```
#include <stdio.h>

int main()
{
    long long n, temp;
    int choice, digit;
    int freq[10] = {0};
    int unique = 1;
    int i;

    printf("Enter a positive integer: ");
    scanf("%lld", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n----- MENU ----- \n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit (0-9)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice)
    {
```

```

case 1:
    while (temp > 0)
    {
        digit = temp % 10;
        freq[digit]++;

        if (freq[digit] > 1)
        {
            unique = 0;
            break;
        }

        temp = temp / 10;
    }

    if (unique == 1)
        printf("Unique\n");
    else
        printf("Not Unique\n");

    break;

case 2:
    while (temp > 0)
    {
        digit = temp % 10;
        freq[digit]++;
        temp = temp / 10;
    }

    printf("\nFrequency of digits:\n");

    for (i = 0; i <= 9; i++)
    {
        printf("freq_%d = %d\n", i, freq[i]);
    }

    break;

default:
    printf("Invalid choice\n");
}

return 0;
}

```

```
UW PICO 5.09 File: day14.c Modified
#include <stdio.h>

int main()
{
    long long n, temp;
    int choice, digit;
    int freq[10] = {0};
    int unique = 1;
    int i;

    printf("Enter a positive integer: ");
    scanf("%lld", &n);

    if (n <= 0)
    {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n---- MENU ----\n");
    printf("1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit (0-9)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = n;

    switch (choice)
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;

                if (freq[digit] > 1)
                {
                    unique = 0;
                    break;
                }

                temp = temp / 10;
            }

            if (unique == 1)
                printf("Unique\n");
            else
                printf("Not Unique\n");

            break;

        case 2:
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            printf("\nFrequency of digits:\n");
            for (i = 0; i <= 9; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

GH Get Help GH WriteOut GH Read File GH Prev Pg GH Cut Text GH Cur Pos  
EX Exit JO Justify WH Where Is NY Next Pg NU UnCut Text NT To Spell

```
UW PICO 5.09 File: day14.c Modified

            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;

                if (freq[digit] > 1)
                {
                    unique = 0;
                    break;
                }

                temp = temp / 10;
            }

            if (unique == 1)
                printf("Unique\n");
            else
                printf("Not Unique\n");

            break;

        case 2:
            while (temp > 0)
            {
                digit = temp % 10;
                freq[digit]++;
                temp = temp / 10;
            }

            printf("\nFrequency of digits:\n");
            for (i = 0; i <= 9; i++)
            {
                printf("freq_%d = %d\n", i, freq[i]);
            }

            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

GH Get Help GH WriteOut GH Read File GH Prev Pg GH Cut Text GH Cur Pos  
EX Exit JO Justify WH Where Is NY Next Pg NU UnCut Text NT To Spell

## 4. Compilation and Execution

```
[aaryanrajput@aaryans-macbook ~ % nano day14.c
[aaryanrajput@aaryans-macbook ~ % clang day14.c -o day14.c
[aaryanrajput@aaryans-macbook ~ % ./day14.c
Enter a positive integer: 12345

----- MENU -----
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice: 1
Unique
[aaryanrajput@aaryans-macbook ~ % ./day14.c
Enter a positive integer: 1122334455

----- MENU -----
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice: 1
Not Unique
[aaryanrajput@aaryans-macbook ~ % ./day14.c
Enter a positive integer: 11223344

----- MENU -----
1. Check whether all digits are unique
2. Find frequency of each digit (0-9)
Enter your choice: 2

Frequency of digits:
freq_0 = 0
freq_1 = 2
freq_2 = 2
freq_3 = 2
freq_4 = 2
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
aaryanrajput@aaryans-macbook ~ % █
```

## Conclusion

The C program **compiled and executed successfully**. Both menu operations were tested, and the actual outputs matched the expected outputs for the prepared test cases.